

Literature-Implied Answer to the Measure-Algebra Question in arXiv:1209.4294

Run `fa_banach_001`, lane 3

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Status

`literature_implied_answer` (full answer to Question 6.8 subquestion).

Source Question

The source paper is P. Koszmider, *Universal objects and associations between classes of Banach spaces and classes of compact spaces*, arXiv:1209.4294.

On page 18, Question 6.8 asks:

Suppose that K is the Stone space of the measure algebra, i.e. the Boolean algebra of Borel subsets of $[0, 1]$ modulo Lebesgue null sets. Is it consistent that $C(K)$ does not isometrically embed into ℓ_∞/c_0 ?

The surrounding paragraph recalls that Dow–Hart proved the corresponding Stone space may consistently fail to be a continuous image of $\mathbb{N}^*$, and then asks whether the Banach-space isometric embedding can also consistently fail.

Supporting Literature

The supporting paper is C. Brech and P. Koszmider, *An isometrically universal Banach space induced by a non-universal Boolean algebra*, arXiv:1505.04776.

On page 2, its introduction says that Dow–Hart proved under OCA that the measure algebra M does not embed into $\wp(\mathbb{N})/\text{Fin}$, while it is well known that

$$C(K_M) \equiv L_\infty \quad \text{isometrically embeds into} \quad C(K_{\wp(\mathbb{N})/\text{Fin}}) \equiv \ell_\infty/c_0,$$

citing Amir–Arbel.

Conclusion

This gives a negative answer in ZFC to Question 6.8 of arXiv:1209.4294. The consistency failure asked for there cannot occur, because the relevant isometric embedding always exists:

$$C(K_M) \hookrightarrow \ell_\infty/c_0$$

linearly and isometrically.

Scope

This packet answers only the measure-algebra isometric-embedding subquestion. It does not answer the broader ZFC equivalence questions from the same survey about universal Banach spaces, isometrically universal Banach spaces, and universal compact spaces.

Search Notes

The run indexes were searched for arXiv:1209.4294, `measure algebra`, `L_infty`, and `l_infty/c_0`. No prior packet for this subquestion was found. The decisive supporting statement was found by local source search in arXiv:1505.04776.

References

- [1] P. Koszmider, *Universal objects and associations between classes of Banach spaces and classes of compact spaces*, arXiv:1209.4294.
- [2] C. Brech and P. Koszmider, *An isometrically universal Banach space induced by a non-universal Boolean algebra*, arXiv:1505.04776.
- [3] D. Amir and B. Arbel, *Continuous functions spaces with the separable complementation property*, Israel J. Math. 29 (1978), 294–306.